

HealthConnect Clinic Experience Lab

Week 4 — Initial Analysis Document (Data Analytics Track)

AnalystLab Africa Experience Lab Internship Program

1. Dataset Overview

The HealthConnect Appointment Dataset contains 5,000 fictional, anonymized appointment records across 1,696 unique patients (an average of roughly 2.9 appointments per patient, ranging from 1 to 9). Each record is described by 18 variables spanning five categories:

- Patient demographics: gender, age, age group
- Appointment details: appointment type, appointment day, appointment time
- Booking information: booking date, appointment date, booking lead days
- Patient history: previous appointments, previous no shows
- Reminder, access and outcome data: reminder sent, reminder channel, distance_to_clinic_km, waiting_time_minutes, appointment outcome

Appointment dates span 1 January 2025 to 30 June 2026, with bookings placed as early as 7 November 2024. The outcome variable (appointment outcome) splits as follows:

Outcome	Count	% of appointments
Attended	2,314	46.3%
No-Show	2,423	48.5%
Cancelled	263	5.3%

Nearly half of all scheduled appointments end in a no-show, confirming this is a substantial and worthwhile problem to investigate further.

2. Data Quality Assessment

2.1 Completeness

Field	Missing values	% missing	Assessment
reminder channel	1,366	27.3%	Expected — matches reminder sent = 'No' exactly (1,366 records). Logical, not a defect; should not be imputed.
distance_to_clinic_km	90	1.8%	Minor, appears randomly distributed across outcomes. Needs a handling decision (exclude vs. impute).
waiting_time_minutes	60	1.2%	Minor, randomly distributed. See anomaly noted below.
All other fields	0	0%	Complete.

2.2 Uniqueness and consistency

- No duplicate appointment_id values and no fully duplicate rows across all 5,000 records.
- previous_no_shows never exceeds previous appointments in any record (0 violations): the historical fields are internally consistent.

- appointment date is never earlier than booking date, and the stated booking_lead_days matches the calculated gap between the two dates in every record: no inconsistencies found.
- age is bounded 18–80, consistent with the dictionary's 'adult patients only' note.
- Categorical fields (gender, appointment type, appointment day, appointment time, reminder sent/channel, appointment outcome) contain only the categories described in the Data Dictionary, no stray or unexpected values.

2.3 Notable anomaly to flag

waiting_time_minutes is populated for No-Show (2,386 of 2,423) and Cancelled (261 of 263) appointments, not only Attended ones, and the average waiting time is nearly identical across all three outcomes (~24 minutes). Since a patient who did not attend could not have logged a real waiting time, this field is likely a simulated/pre-assigned value rather than an attendance-linked measurement. It should be treated with caution or scoped only to Attended records in later analysis rather than assumed to be causally linked to no-shows.

3. Relevant Business Questions

- Q1. Which patient, booking, and appointment characteristics are most strongly associated with no-shows?
- Q2. Does sending an appointment reminder, and the channel used measurably influence attendance?
- Q3. How much appointment capacity is lost to no-shows and cancellations, and where does this loss concentrate (by day, time slot, or appointment type)?
- Q4. Are patients with a prior history of missed appointments more likely to miss future ones, and can this history be used to flag higher-risk bookings?
- Q5. How does distance to the clinic and booking lead time relate to attendance behavior?

4. Potential KPIs

At this stage, the KPIs below are identified and justified only, none have been fully calculated, analyzed, or visualized, per the Week 4 scope.

KPI	Definition	Linked question
No-Show Rate (overall & by segment)	% of scheduled appointments with outcome = No-Show, segmented by appointment type, appointment day, appointment time	Q1, Q3
Reminder Effectiveness	Attendance rate compared across reminder sent = Yes vs. No, and across reminder channel	Q2
Lost Slot Rate	% of scheduled appointments resulting in No-Show or Cancelled (capacity lost to non-attendance)	Q3
Repeat No-Show Rate	No-show rate for patients with previous_no_shows > 0 vs. previous_no_shows = 0	Q4
Distance/Lead-Time Banded Attendance Rate	Attendance rate across distance_to_clinic_km bands and booking_lead_days bands	Q5

An early cut of the data already suggests previous_no_shows is a meaningfully strong signal: patients with at least one prior no-show show a no-show rate of 55.4% on their current appointment, versus 43.5% for patients with a clean history, an early candidate for the strongest predictor to explore further.

5. Initial Analysis Approach

- Phase 1 Data preparation in Power Query: import the appointment data, confirm data types (dates, integers, decimals) match the Data Dictionary, and apply the null-handling decision for distance_to_clinic_km and waiting_time_minutes (likely exclusion or a simple substitution, given they're under 2% of records).

- Phase 2 Modelling in Power BI: load the cleaned table, build supporting date and age-band dimension tables if needed, and confirm relationships/grain (one row per appointment).
- Phase 3 Univariate and bivariate exploration: distribution visuals for each candidate variable, then appointment_outcome broken out by each variable (matrix/stacked bar visuals) to surface the strongest associations, extending the early cross-tabs in Section 4.
- Phase 4 KPI calculation with DAX: build measures for the five KPIs above (No-Show Rate, Reminder Effectiveness, Lost Slot Rate, Repeat No-Show Rate, Distance/Lead-Time Banded Attendance Rate) once business questions are prioritized with stakeholders.
- Phase 5 Dashboard build: assemble the measures and visuals into an interactive Power BI report with slicers (appointment_type, day, time, age group) so clinic decision-makers can filter no-show patterns themselves.

Tools: Power Query for data import, cleaning and shaping; Power BI (data model + DAX measures + report visuals) for analysis and the interactive dashboard.

6. Assumptions, Limitations, Risks and Dependencies

6.1 Assumptions

- The dataset is fictional and synthetic; patterns found are for learning purposes and may not mirror a real clinic's dynamics, though the structure is realistic enough to practice on.
- reminder_channel nulls are assumed to mean 'no reminder sent' rather than a data capture failure, based on the exact match with reminder_sent = No.

6.2 Limitations

- waiting_time_minutes cannot be trusted as a genuine attendance-linked measurement (see Section 2.3) and should be used cautiously or excluded from no-show driver analysis.
- No direct cost or financial-impact field exists, so any 'cost of no-shows' framing in later weeks will need an assumed value per slot.
- appointment_outcome bundles distinct behaviors: a Cancelled appointment (patient gave notice) is operationally different from a No-Show (no notice), KPI definitions need to treat these separately rather than merge them by default.

6.3 Risks and dependencies

- Decisions on null-handling (distance, waiting time) made in Week 5 will affect downstream KPI values — these should be documented, not silently changed.
- Later weeks depend on the Data Dictionary remaining the single source of truth for field definitions; any relabeling of appointment_outcome categories would require re-running this quality assessment.
- If subsequent tracks (Data Science / ML Engineering) rely on this same dataset, the anomaly in waiting_time_minutes should be communicated to avoid it being used as a predictive feature without caveat.